

Intelligent Chatbot Assistant in Agriculture Domain

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Abstract. Agriculture is known as the economic game changer of India. It is the primary driver of GDP growth because of India's robust agricultural industry, and proper knowledge about agriculture techniques helps increase crop yield. So, answering the different types of crop-related queries are very important. We proposed the intelligent chatbot application in the agriculture domain so that farmers can get the correct information about farming practices. Our system is farmer-friendly and capable enough to instantly answer farm-related queries from the knowledge base, such as plant protection, fertilizer uses, government schemes, and many others. We used the agriculture-related data in question-answer format and implemented the pre-trained model of the Sentence-Transformer approach in our chatbot application to answer providing. We also deployed the TF-IDF and Bag-of-Words method but achieved a 96% accuracy rate for the test data in the sentence transformer pre-trained model. With the help of API services, our system also shows the crop's latest mandi (market) rate and current weather information. So, the proposed chatbot system will keep the contribution for farmer's cost savings. Overall, our chatbot system is straightforward and more efficient for the farmer to make better decisions.

Keywords: Sentence Transformer Pre-Trained Model · TF-IDF · Bag of Words · Pegasus Model · Mandi(Market) Rate API · Cosine Similarity · Weather API.

1 Introduction

In Asia, most countries depend on agriculture to fulfill the food demand of the people, and agriculture is a significant contributor to the country's increasing productivity. In India, nearly 60% of the population works in agriculture [1], and lots of people from rural areas are farming for their livelihood as their occupation [2]. However, most Indian farmers do their farming process in their traditional method, and because of the technological or communication gaps, they are unknown of the latest farming information. Most of the time, they are unable to solve farming-related problems. As a result, they experience limited crop growth, which causes them to incur losses.

So, to solve the agriculture sector issues, the government already take many necessary steps. The farming authority also organizes several programs for farmers, like Front Line Demonstrations with the help of a network of Krishi Vigyan Kendras in every district, Rashtriya Krishi Vikas Yojana, and other services [3]. Moreover, we can see many existing services for farmers, like the Kisan calling center, eNam, and farmers portal.

The Ministry of Agriculture and Farmers Welfare is continuing the Kisan call center to facilitate efficient communication of farmer’s queries in various languages. However, due to many incoming calls, it is sometimes challenging to attend to all of them promptly, and users have to wait a long time to ask their queries. As a result, users need help getting the solution to their query. Besides, Agriculture Ministry has also introduced a web-based service called the Farmers Portal, which offers a comprehensive range of information and services to farmers in one convenient site. The main feature of this digital platform is agri-Advisory, crop management, animal husbandry management, and so on [4]. Despite its benefits, the Farmers Portal has some limitations, including fixed support, and users may need to have the technical knowledge to navigate the platform. These challenges may be complex for some users to access required agricultural information. Moreover, we can see the eNAM - National Agriculture Market platform for selling the farmer produce, the Kisan Suvidha application, which integrates all the farm-related services.

Most of the existing works used the traditional approach or pre-implemented platform to deploy the farmer chatbot applications, which is less suitable for a large number of chatbot knowledge. We can also see the most recently introduced technique, CHATGPT, which can answer all types of questions. However, in the agriculture domain, some factors also limit this application. We introduced an intelligent chatbot system for farmers to get suitable solutions for various farming practice queries. We aim to enhance farmer operations and eventually support the agriculture industry’s expansion and sustainability. One of the biggest challenges of deploying an agriculture chatbot application is ensuring reliable and efficient data. The collected data for the agriculture chatbot application must be relevant, accurate, and up-to-date to provide the farmer with the best possible answer and support. To overcome this challenge, we collected data from KCC (Krishi Vigyan Kendra) and focused on manual data collection, knowledge models, and real-time API information. In our chatbot, users can get answers like plant protection, cultural practices, fertilizer uses, water management, and so on. Our system also provides the current market price of different crops in India, real-time weather information through the API system, and supports the Hindi and English languages. We developed our application in the Indian context, so we consider the Hindi language as it is the official language of India. We deployed a sentence transformer, the latest AI technique, and built a user-friendly system for non-technical individuals. Our chatbot is convenient and provides customized information about farming-related user queries.

The paper organizes as follows: Section II discusses the related works. Section III describes the proposed methodology, and IV explains the dataset collection and

preprocessing. Section V describes the dataset analysis. Section VI discusses the approaches, and Section VII evaluates the result and discussion. Section VIII concludes the overall work.

2 Related Works

Bhardwaj et al. [5] proposed the farmer-assistive chatbot using the KNN and the sequence-to-sequence model for the answer generation that is multi-language supported. Arora et al. [6] implemented the RNN seq2seq technique for the conversational system using the KCC and web scraping data. They also used the CNN approach to detect crop disease, and the weather information will come through the API services. Niranjana et al. [7] discussed the overall survey on the chatbot application. They described the shortcomings of the various technique and suggested the seq2seq as a good solution in the case of the chatbot assistant. Gounder et al. [8] and Mohapatra et al. [9] implemented the TF-IDF approach to generate the vector and used the cosine similarity method to find the similarities of the question from the KCC dataset. Besides, they considered the lemmatization process instead of the stemming process for the data preprocessing and deployed the application in Android [8]. On the other hand, they [9] used MongoDB as a database and found an excellent similarity score for the query text's more matching words.

Nayak et al. [10] introduced the agriculture expert chatbot application using the chatterbot library, where they collected agriculture-related data in question-answer format. They used the Levenshtein distance and the best-match approach to answer the user queries correctly. Here, users can also ask their query through voice and get agro-expert support in case of chatbot is unable to answer.

Jain et al. [11] deployed the FarmChat android application with audio and (audio+text) and compared the output efficiency in their study. Here, they considered not only the Kisan calling center data but also collected information from local cultivators and agriculturalists so that the application could answer farm-related questions about potato crops easily. Thatipelli et al. [12] designed an Agriculture robot with different types of sensors for collecting data from the farmland and storing it in the IBM cloud for real-time analysis. Besides, they also deployed the chatbot based on the collected data, which suggests a suitable fertilizer considering different parameters and detects plant diseases. Momaya et al. [13] also introduced the farmer assistive chatbot name "Krushi" using the RASA X framework, which is able to provide almost 70% of query answers.

Gunawan et al. [14] proposed an IoT - based chatbot application. With the help of an IoT device, it senses the field humidity, temperature, and soil moisture and sends the sensor data to the database regular basis. The chatbot uses this data to provide real-time information and help urban farmers to improve agriculture's efficiency. Mostaco et al. [15] proposed the AgronomoBot, which can also answer the farmer's queries from the agriculture wireless sensor network data. The data will come from the WSN database and deploy the application in the telegram for the user interface. Kiruthika et al. [16] proposed a quite different chatbot system

for the consumer and the farmers, where a farmer can sell their crops at a suitable price and get more profit. Their proposed approach’s main functionalities are collecting the farmer details and crop-related information, recommending farmers as the user needs, and constantly updating the farmer crops details.

Shao et al. [17] proposed a neural network-based transformer to solve the tasks of answer selection in the question-and-answer systems. They implemented the (BiLSTM) bidirectional long short-term memory, transformer-based, and prioritized sentence embedding tasks. Here, Ngai et al. [18] deployed different transformer-based models like BERT, ALBERT, and T5 for Question Answering (QA) systems on COVID-19. In the result analysis, they achieved the effective F1 score for the BERT and a better exact match for the ALBERT technique. Soldaini et al. [19] proposed another answer selection approach, named cascade transformer. On several benchmark datasets, they evaluated the Cascade Transformer, and the experiment’s outcomes indicate that the Cascade Transformer is a promising method for choosing answer sentences in QA applications.

3 Proposed Methodology

We proposed an effective chatbot application to answer farm-related queries using state-of-the-art technology. For the agriculture chatbot application, the dataset plays a significant role. We combined various sources of data: the Kisan Calling Center dataset, the Knowledge model data, and others in the question-answer structure. We preprocessed the data using tokenization, stemming, text normalization, and other preprocessing techniques, imported the necessary library as model requirements, and loaded the pre-trained model. The sentence transformer pre-trained model encoded the input query and the knowledge base questions and computed similarity scores using the cosine similarity approach. We set a threshold value based on the chatbot’s performance to ensure the reliability of the similarity score. Moreover, if the user wants to know the Weather Information or Mandi Rate, our chatbot will display the information through the API services regarding their search. For the weather condition, users need to input the location name, like city, and to get mandi rates with more information; they must search by crop name. Here, we used the spelling checker library so that if the user inputs the wrong spelling of the crop’s name, then it will take it as the correct one. As we already said, our system is user-friendly, meaning it will also answer the query in Hindi. We deployed the google translate API for Hindi language translation. So, when users ask a query in Hindi, it will translate into English, find a similar question to answer and deliver it in Hindi. In the chatbot, users may ask a question differently. So, to analyze the results, we used the Pegasus model to generate paraphrases of the test questions and checked whether the model identified the correct answer or not. We implemented different approaches but found that the sentence-transformer pre-trained model provided the best results for the test dataset. Undoubtedly, we achieved an excellent performance rate with the sentence transformer pre-trained model. In figure 1, we can see the proposed architecture of our proposed work.

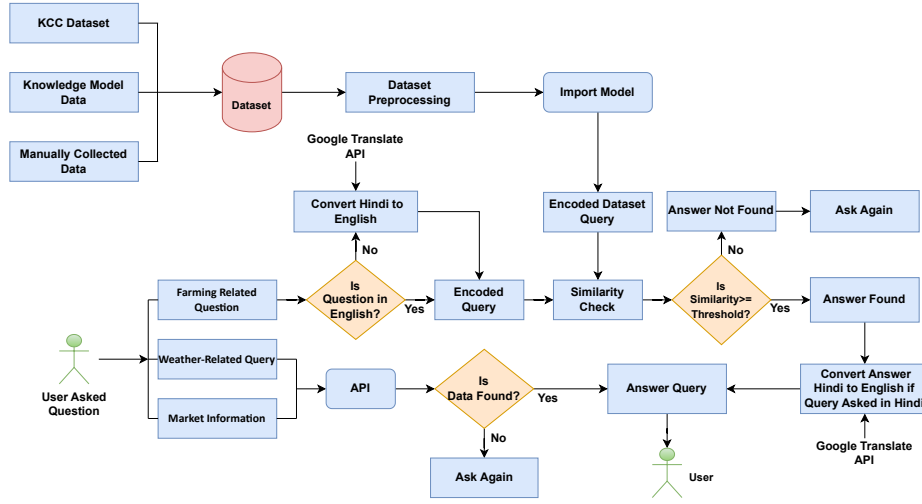


Fig. 1: Proposed Architecture

4 Dataset Collection and Preprocessing

The utilization of datasets is a crucial part of developing question-answering chatbots. We collected the dataset from different sources and preprocessed it to build our farmer chatbot system. In this section, we will discuss it more elaborately.

4.1 KCC (Kisan Calling Center)

Earlier, we discussed the Kisan calling center, a help center facility for the farmer. A total of 21 Kisan call center locations exist in different states of India. This kind of service helps the farmer acquire more knowledge about farming. For the farmer chatbot application in Indian Region, the KCC dataset is essential. This open-source dataset is available on the government website [20] (data.gov.in). So we collected the most recent KCC data of Punjab, Tamil Nadu, and Maharashtra for a different district. We have 65.11% for the Punjab state, Maharashtra 22.39%, Tamil Nadu 12.48%, and more than fifteen thousand total KCC raw data. The dataset has nine features: Seasons, Sectors, Crops, Query Type, Query Text, Answer, StateName, District Name, and CreatedOn. However, in this case, we consider only the valuable feature that contributes most to the generation of answers in our chatbot application.

- **Category:** This column represents the category of the query asked by the users, like vegetables, cereals, etc.
- **Crop:** This column represents the specific crop mentioned in the user's query.
- **Query Type:** This column represents the type of query asked by the users.

- **Query Text:** This column represents the actual text of the query asked by the users.
- **KCC Answer:** This column represents the answer given by the Kisan Calling Center representative to the user’s query.

Data Preprocessing After the dataset collection, it is essential to preprocess the data. The KCC dataset was in JSON format, and we converted it into CSV format to understand the data more efficiently. We merged all the district-wise KCC collected data and evaluated it for further procedure. After analyzing the KCC data, we found lots of duplicate data, spelling mistakes, incomplete sentences, etc. We followed the remaining steps to preprocess the KCC datasets.

- **Used Google Translate API:** The query answers of the KCC dataset were in the local language. To build our system, we considered only the English language. So, we translated the local language to English for better understanding and efficiency in working with the dataset.
- **Dropped the Weather-Related Rows:** Most KCC dataset queries were related to weather conditions. We dropped the weather-related query almost 30% of the total KCC data because the user will get the answer through the weather API services.
- **Dropped the Contact Number Asked Rows:** We found 7% data related to asking for the contact number details of Krishi Vigyan Kendra, agriculture institution, and nodal officer details. However, there needed to be more data, and besides, most of the government scheme-related query answers were the nearest agriculture center contact number. This kind of answer may not be helpful to the user. So, we did not count it because we collected the contact details manually for all asked places in the KCC dataset query and added more government scheme data with suitable answers.
- **Dropped the Duplicates Rows:** We also found lots of duplicate data, which is 40% in the KCC datasets. So, we dropped the duplicates.
- **Dropped Animal and Flowers Related Data:** We did not count the animal and flower-related data in our KCC data consideration. However, in the future, we plan to incorporate this information to enhance the knowledge of our chatbot.
- **Dropped the Farming Fair Information:** In analyzing the Kisan Calling Center dataset, we came across numerous queries related to agriculture fair information. We realized that the queries were referring to fairs holding dates that took place in the past years, and the answer was no longer valid. Therefore, we excluded such queries from our dataset.
- **Dropped the Market Information of Crop Price:** We dropped the market information-related data by almost 1.48% because our chatbot system will provide the information regarding the market crop price through the API services.
- **Others:** After completing the aforementioned steps, we noticed that the dataset had many incomplete answers, inappropriate information, null value rows, and insufficient crop data. Additionally, many repeated questions with

different structures and answers could cause problems evaluating the model's performance. So we removed these kinds of information and decided to focus on a limited number of crops with sufficient information available to improve the dataset's overall quality and ensure the effectiveness of the chatbot application.

After the KCC data cleaning, it is quite understandable that the considered data is not sufficient enough for our proposed chatbot application. In the next upcoming sections, we will discuss our data collection other procedures, and API data.

4.2 API Data

In the agriculture sector, knowing about the recent update on weather conditions and the crop price in a different market is essential so that farmers can make immediate decisions. As discussed earlier, we aim to bring all the farming-related functionalities into our chatbot application. So for that, we also deployed the following:

- **Weather Condition**
- **Mandi Rates of Various Commodities**

With the help of open weather API, we displayed the real-time weather parameter like temperature, humidity, weather report, and wind speed of the searching place [21]. It will help farmers with efficient planning and better resource management. The second is the "Current Daily Price of Various Commodities from Various Markets or Mandi." The API services are available on the [20] (data.gov.in) website, which is open-source for all. This API provides ten features like state, district, market Name, crop name, variety, grade, data arrival dates, crops minimum, maximum, and model price (Rs./Quintal). In our chatbot application, users only have to input the commodities names; according to that, our chatbot will show the Mandi Rates with state, district, and Market names. Here, we also deployed the spell-checker functionalities in the commodities search to keep in mind that if the user makes mistakes in writing the crop name, it will automatically take the correct spelling.

4.3 Knowledge Model

In the chatbot application, the completeness of data plays a significant role because it will help to enlarge the knowledge and enable the application to answer different crop-related questions. We all know that most people in India depend on rice and wheat crops. However, we found a lack of information on disease protection, weed management, and water management for these crops in the KCC dataset. To address this gap and enlarge the chatbot's knowledge, we collected data from agricultural experts on rice, wheat, and maize crops. Although the data was not in a question-and-answer format, it contained valuable information on crop varieties, sowing times, disease symptoms, protection

procedures, nutrient management, field preparation, fertilizer, and water management. It is necessary to the conceptual representation of the data and its relationships to understand the data completeness or required information. In that case, the knowledge model is more adaptable and scalable. It provides a visual understanding and clearly and consistently represents structured data experience. The flexibility of the knowledge models helps to accumulate information in an organized manner and create better decisions. We took inspiration from the Agropedia source [22] to design the knowledge model. In this work, we focused on the knowledge model of agriculture for rice, wheat, and maize crops. These models focused on nutrient management, environmental requirement, varieties, field preparation, sowing time, soil management, fertilizers, water management, weed management, pest management, and diseases management. So these knowledge models helped us to comprehend data completeness, identify the missing information, and understand the relationship. Overall, we found the highest number of rice data, and maize is in the last position. We converted this data into a question-answer format that can enhance the efficiency of chatbot applications and contribute towards farmer’s adoption of successful as well as productive agricultural practices.

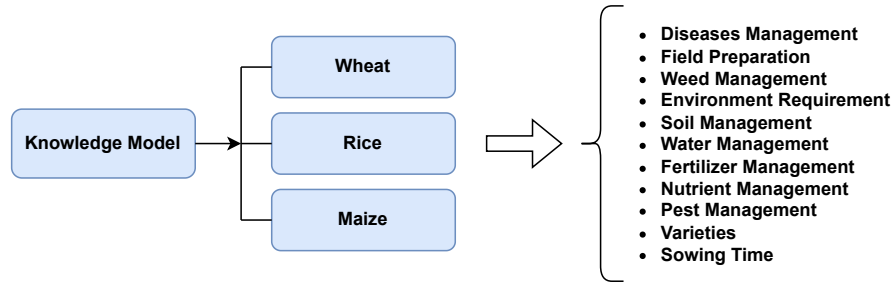


Fig. 2: Different Concepts of Rice, Wheat, and Maize Knowledge Model

4.4 Manually Data Collection

To make our chatbot application more efficient and a better resource, we took help from various governments and farming-related sources [23–29] for required query type data. So we manually created different questions regarding the required data and stored the information in the answer structure. To cater to the farmer’s diverse information needs, we prioritized the various query types for each crop, ensuring that a range of information will be available. We also included frequently asked questions about government schemes, such as how to apply, required documents, benefits, and other necessary information. Moreover, in the previous section, we discussed many queries about the Krishi Vigyan Kendra, agriculture institution, and Nodal officer contact details. So, we added

the contact number of Krishi Vigyan Kendra all over India with senior scientist details, more than 60 agriculture institutions, and nodal officer information. Most of the contact details data was in the tables layout in the government sources. We arranged the collected information in the question-answer structure to help the chatbot system provide a suitable answer to the user.

5 Dataset Analysis

After collecting the data, we have five columns: Category, Items, Query Type, Query, and Answer. Here the item part includes the different crop’s names, the government scheme, and the contact number of the farmer helping center/officer.

Table 1: Analysis of Crop’s Category and Query Type

Category	Total Data(%)	Query Type	Total Data(%)
Vegetables	25.84%	Plant Protection	27.38%
Cereals	25.50%	Cultural Practices	21.14%
Fruits	11.31%	Varieties	15.16%
Others	10.58%	Others	10.58%
Pulses	10.17%	Nutrient	7.36%
Oilseeds	6.07%	Fertilizer Uses	6.49%
Spices	5.55%	Seeds	4.61%
Schemes	2.59%	Weed Management	2.38%
Fiber Crops	2.30%	Government Schemes	2.69%

Table 1 shows the category column information and the count of different agricultural query types. The largest segment of the table represents the vegetable category, almost 25.84 % of the total count—the next largest segment for the Cereals category, followed by the fruits. The remaining categories, such as Schemes, Pulses, and others, would occupy smaller segments proportional to their counts. In Table 1, the most common query type is plant protection, with a count of 27.38%, followed by cultural practices, with 21.14% queries. These two query types represent over 48% of the total queries. The next is Varieties, with a count of 15.16%, followed by others query types, with an exact count of 10.58%.

Besides, we have a vast number of rice data, with a 14% count representing the highest crop data. The next largest crop is Wheat, with a count of 6.35%, followed by maize and Bengal gram. Other crops, such as Potato, Brinjals, Tomato, Garlic, and Cauliflower, are also in there but to a lesser extent. The remaining crops, such as Coriander, Mustard, Chillies, Turmeric, and others, represent a smaller proportion of the total crops. Besides, the dataset contains 496 contact details for KVK centers and information about various government schemes, such as Pradhan Mantri KISAN Samman Nidhi Yojana and the PM

Kisan credit card scheme. Moreover, the dataset also contains the contact details of agriculture institutions, universities, and Nodal Officers.

The main challenge in the farmer’s question-answer chatbot application is collecting accurate and suitable data. Besides the Kisan calling center data, we also included the missing information from scratch, making the system more efficient for the farmer. Overall, there are more than six thousand total data. The analysis shows that the chatbot application has sufficient query type knowledge for the limited crop, government scheme, and contact number details to enhance the chatbot’s ability to understand and respond accurately to user queries.

6 Models

For deployed the chatbot application, we used three approaches: Bag-of-Words, TF-IDF, and a Pre-trained model of the sentence transformer. These methods are famous and can see the contribution to NLP tasks, including intent classification and entity recognition. Bag-of-words is a popular approach for representing text data as numerical vectors. Depending on the specific task, it is traditional and valuable. In our chatbot application, we used the BoW to generate the feature vector of the questions. Secondly, the widely used method in NLP is called TF-IDF (Term Frequency-Inverse Document Frequency). Unlike BoW, the TF-IDF prioritizes the significant word. In our chatbot, we also used the TF-IDF technique to generate the feature vector of the queries.

Finally, Sentence Transformer is the latest approach in NLP that provides pre-trained models for various uses. Here, we used the quite effective, more petite, faster pre-trained model "all-MiniLM-L12-v2" and retained a high level of accuracy as well as good performance [30]. SentenceTransformer models have undergone pre-training on extensive textual data to enhance their performance on particular tasks. We used the pre-trained model for encoding text into vector representations and generating sentence encoding involves several steps. The model first tokenizes the input sentence by splitting it into individual words. Then this model also integrates pooling and other processes in the encoding phase. The model can understand the connections between phrases and words and their context in a sentence through the pre-training procedure. The pre-trained sentence transformer model encoded knowledge-base questions and the user-asking query in our application. The highest cosine similarity score counted to determine the most similar question from the knowledge base for asking the query. Through the testing part, we noticed that this pre-trained model also performed better in the case of synonym words in a sentence and took less time to provide the answer.

7 Result and Observations

After implementing the chatbot application, it is mandatory to check whether it will provide the correct answer. So, we randomly selected unique 100 and 200 questions from our original datasets as test data with the correct answer.

After that, we deployed the pre-trained pegasus model for both test datasets to generate the paraphrase of these questions to analyze the similarity model performance. Pegasus is a transformer-based language model used for the NLP tasks like summarization [31]. We used the pre-trained pegasus model to produce paraphrased phrases based on its internal understanding of the input text. Human evaluation is essential in machine-generated text, so we manually evaluated the pre-trained pegasus model [32] [33]. This process is time-consuming but flexible. It also helped us understand the paraphrased sentence’s meaning and whether it was relevant to the original sentence or not. We checked different facts in each of the paraphrased sentences, such as Meaning, Fluency, and Relevance. We observed that most of the paraphrased questions semantically connected with the original dataset questions, and some were incomplete, such as the actual questions that correctly defined the crop variety, like “Information Regarding Soybean variety SL 958 (2014)?” but in the paraphrased sentence “Is there any information soybean variety SL 958?” and for asking contact number or address of a place like Kisan Vigyan Kendra, the paraphrased question missed the state or district name of KVK. We also counted these questions because, in the chatbot application, users can ask queries in different ways. So this helped us evaluate the system’s effectiveness.

We considered the accuracy, average cosine similarity score, and the threshold value for evaluating all three similarity models. The threshold performs as a required minimum similarity score before a question can be considered a match. We compared the predicted answer for each question with the actual answer for accuracy count. A higher accuracy signifies that the system works effectively, identifying the questions most similar to each other and providing the correct answers for most inquiries.

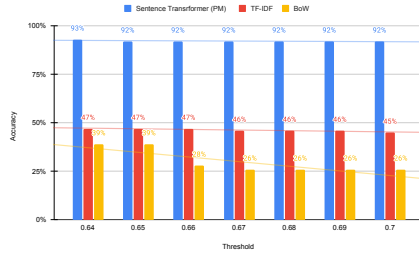


Fig. 3: Test Accuracy of 100 Questions

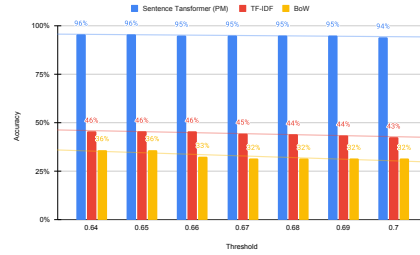


Fig. 4: Test Accuracy of 200 Questions

Figure 3 and 4 show the accuracy of the Sentence Transformer pre-trained model, TF-IDF, and BoW for 100 and 200 test questions. Here, we considered the different threshold values (0.64 to 0.70) for the model’s performance analysis. We defined the threshold value in our chatbot application based on the excellent output. The Sentence Transformer pre-trained model (PM) has the highest accuracy of 93% for a 0.64 threshold value. The TF-IDF model has an

accuracy range from 47% to 45%. This model is less accurate than the Sentence Transformer (PM) model but performs better than the BoW model. The BoW model has an accuracy range from 39% to 26%. This one is the least accurate among the three models and outperformed compared to the others. As seen from the figure 4 for 200 questions, the Sentence Transformer (PM) has the highest accuracy, followed by TF-IDF and BoW. The accuracy of the models decreases as the threshold increases. At the threshold of 0.64, the Sentence Transformer (PM) has an accuracy of 96%, TF-IDF has 46%, and BoW has 36%. When the threshold goes to 0.7, all approaches' accuracy decreases. The accuracy results may vary based on the dataset and the asked questions. The threshold value is also an essential factor in determining the model's accuracy, and a higher threshold value can result in lower accuracy.

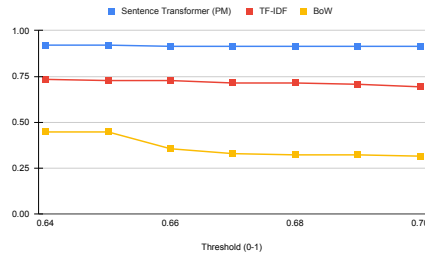


Fig. 5: Average Cosine Similarity Score for 100 Questions for All Three Models

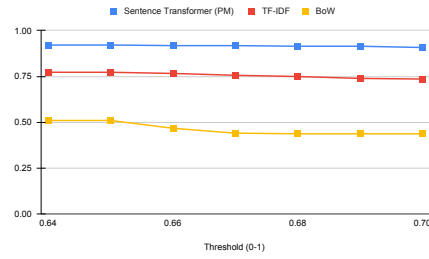


Fig. 6: Average Cosine Similarity Score for 200 Questions for All Three Models

In NLP, the similarity-matching approach is well-known and determines the similarity between two texts or documents. We calculated the average cosine similarity by summing up the similarity scores of the highest-scoring question, which also satisfied the threshold value, and then dividing that by the number of total questions. For the 100 questions, in 5, the Sentence Transformer method consistently has a better cosine similarity value than the other two methods, with a score of 0.9139 to 0.9205 across all threshold values. The Sentence Transformer method performed better than the other two methods, with higher cosine similarity scores across all threshold values. For the 200 questions, in 6, we can see that the Sentence Transformer model has the highest average cosine similarity score, followed by the TF-IDF and BoW models. At a threshold of 0.64, the Sentence Transformer model has an average cosine similarity score of 0.9211, which is higher than the scores for the TF-IDF model 0.7725 and the BoW model 0.5095. Each model's average cosine similarity score decreases as the threshold value increases. At a threshold of 0.7, the Sentence Transformer model still has the highest average cosine similarity score. The TF-IDF and BoW scores have also decreased but are still lower than the Sentence Transformer model. Overall, we also find the effectiveness of the Sentence Transformer pre-trained model in determining sentence similarity based on the average cosine similarity scores.

Asked Question: Can you tell me about the potato seed treatment?

Question find from Database: Information Regarding seed treatment of potato?

Answer: For seed treatment in potatoes, 2.5 ml of Monseron per liter of water should be mixed and the seeds should be immersed in this solution for 10 minutes.

Example 1. Q&A of the Chatbot Application

Here, we can see the question-answering process of the Sentence Transformer pre-trained model. Our proposed chatbot model is well-organized and straightforward. The sentence transformer pre-trained model is the latest technology and provides adequate performance in the question-answer system. We got 96% accuracy for the 200 and 93% for test question 100 for the sentence transformer (PM), which is the highest compared to the others. Our chatbot is limited in knowledge, meaning it can only answer the database question and supports only Hindi and English. We will enlarge our database to answer all agriculture-related questions in the future and make it available in other languages. Our application is less complex and farmers can make data-driven decisions that lead to improved agricultural production, efficiency, and profitability with the help of a chatbot assistant.

8 Conclusion

The chatbot system is the most extensive creation in Artificial Intelligence because it solves the user problem quickly and positively impacts the world. Our proposed agricultural chatbot system enables immediate responses to farming-related inquiries in Hindi and English. We used the Sentence Transformer pre-trained method, a state-of-the-art technique in NLP, because of large-scale pre-training and high performance. We obtained a higher accuracy rate in the pre-trained model compared to the other two methods. The proposed method can increase farmer's knowledge and help them make informed decisions about their crop management and operations by providing an accessible, user-friendly system with reliable information. Through the API system, the user will receive real-time responses, which will help them stay more current on market prices and weather information. In the future, we will increase the chatbot's knowledge and add more functionalities like suggesting the crop advisor details who is more supportive regarding the queries. We will also add the query suggestions method depending on the user asking the question and make the system robust.

References

1. Statista Research Department Aprajita Minhas. Topic: Agriculture in India. <https://www.statista.com/topics/4868/agricultural-sector-in-india/#topicOverview>, 2023. Accessed: 2023-01-30.

2. Adi(Alternative Development Initiative) International. Topic: Agriculture in India. <https://www.adi-international.org/agriculture-rural-development/>. Accessed: 2023-02-03.
3. Ministry of Agriculture and Farmers Welfare. Increasing Knowledge and Awareness Among Farmers to Enhance the Production and Productivity. <https://pib.gov.in/newsite/PrintRelease.aspx?relid=124565>, 2015. Accessed: 2023-02-03.
4. Department of Agriculture, Cooperation, and Farmers Welfare. Farmer's Portal. <https://farmer.gov.in/FarmerHome.aspx>. Accessed: 2023-02-05.
5. Tanmay Bhardwaj, Prathmesh Deshpande, Tanvi Murke, Shruti Deshpande, and Kashmira Deshpande. Farmer-assistive chatbot in indian context using learning techniques. In *Security Issues and Privacy Threats in Smart Ubiquitous Computing*, pages 239–246. Springer, 2021.
6. Bhavika Arora, Dheeraj Singh Chaudhary, Mahima Satsangi, Mahima Yadav, Lotika Singh, and Prem Sewak Sudhish. Agrirobot: a natural language generative neural networks engine for agricultural applications. In *2020 International Conference on Contemporary Computing and Applications (IC3A)*, pages 28–33. IEEE, 2020.
7. Prashant Y Niranjana, Vijay S Rajpurohit, and Rasika Malgi. A survey on chat-bot system for agriculture domain. In *2019 1st International Conference on Advances in Information Technology (ICAIT)*, pages 99–103. IEEE, 2019.
8. Susheel Gounder, Mahaleppa Patil, Vishal Rokade, and Nikita More. Agrobot: An agricultural advancement to enable smart farm services using nlp. *Journal of Emerging Technologies and Innovative Research*, 2021.
9. Sandeep Kumar Mohapatra and Anamika Upadhyay. Using tf-idf on kisan call centre dataset for obtaining query answers. In *2018 International Conference on Communication, Computing and Internet of Things (IC3IoT)*, pages 479–482. IEEE, 2018.
10. Vandana Nayak, NH Sowmya, et al. Agroxpert-farmer assistant. *Global Transitions Proceedings*, 2(2):506–512, 2021.
11. Mohit Jain, Pratyush Kumar, Ishita Bhansali, Q Vera Liao, Khai Truong, and Shwetak Patel. Farmchat: a conversational agent to answer farmer queries. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2(4):1–22, 2018.
12. Prathyusha Thatipelli and R Sujatha. Smart agricultural robot with real-time data analysis using ibm watson cloud platform. In *Advances in Clean Energy Technologies: Select Proceedings of ICET 2020*, pages 415–427. Springer, 2021.
13. Mihir Momaya, Anjnya Khanna, Jessica Sadavarte, and Manoj Sankhe. Krushi—the farmer chatbot. In *2021 International Conference on Communication information and Computing Technology (ICCICT)*, pages 1–6. IEEE, 2021.
14. Reza Gunawan, Ichan Taufik, Edi Mulyana, Opik T Kurahman, Muhammad Ali Ramdhani, et al. Chatbot application on internet of things (iot) to support smart urban agriculture. In *2019 IEEE 5th International Conference on Wireless and Telematics (ICWT)*, pages 1–6. IEEE, 2019.
15. Gustavo Marques Mostaco, Icaro Ramires Costa De Souza, Leonardo Barreto Campos, and Carlos Eduardo Cugnasca. Agronomobot: a smart answering chatbot applied to agricultural sensor networks. In *14th international conference on precision agriculture*, volume 24, pages 1–13, 2018.
16. Usha Kiruthika, S Kanaga Suba Raja, V Balaji, and CJ Raman. E-agriculture for direct marketing of food crops using chatbots. In *2020 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS)*, pages 1–4. IEEE, 2020.

17. Taihua Shao, Yupu Guo, Honghui Chen, and Zepeng Hao. Transformer-based neural network for answer selection in question answering. *IEEE Access*, 7:26146–26156, 2019.
18. Hillary Ngai, Yoona Park, John Chen, and Mahboobeh Parsapoor. Transformer-based models for question answering on covid19. *arXiv preprint arXiv:2101.11432*, 2021.
19. Luca Soldaini and Alessandro Moschitti. The cascade transformer: an application for efficient answer sentence selection. *arXiv preprint arXiv:2005.02534*, 2020.
20. Ministry of Electronics National Informatics Centre (NIC) and Information Technology. Open Government Data (OGD) Platform India. <https://data.gov.in/>. Accessed: 2023-02-05.
21. Open Weather Map. . <https://openweathermap.org/>. Accessed: 2023-02-05.
22. Agropedia. Knowledge Models. <http://agropedia.iitk.ac.in/content/knowledge-models-0>. Accessed: 2023-02-03.
23. Apni Kheti – Empowering Rural India Digitally. <https://www.apnikheti.com/en/pn/home>. Accessed: 2022-12-18.
24. Punjab INDIA Punjab Agricultural University (PAU) Ludhiana. <https://www.pau.edu/>.
25. New Delhi ICAR-Indian Agricultural Research Institute. <https://iari.res.in/index.php/en/iari-new-delhi>.
26. IIWBR – Indian Institute of Wheat and Barley Research. <https://iiwbr.icar.gov.in/>.
27. Jal Shakti Abhiyan. <https://ejalshakti.gov.in/jsa/>.
28. Direct Benefit Transfer In Agriculture Mechanization. <https://agrimachinery.nic.in/Master/User/StateWiseNoduleOfficerReport>.
29. Vikaspedia Domains. <https://vikaspedia.in/agriculture>.
30. Technische Universität Darmstadt Nils Reimers. Pretrained Models - Sentence-Transformers documentation. https://www.sbert.net/docs/pretrained_models.html. Accessed: 2022-12-12.
31. Pegasus. https://huggingface.co/docs/transformers/model_doc/pegasus. Accessed: 2023-02-03.
32. Daojian Zeng, Haoran Zhang, Lingyun Xiang, Jin Wang, and Guoliang Ji. User-oriented paraphrase generation with keywords controlled network. *IEEE Access*, 7:80542–80551, 2019.
33. Chris van der Lee, Albert Gatt, Emiel van Miltenburg, and Emiel Krahmer. Human evaluation of automatically generated text: Current trends and best practice guidelines. *Computer Speech & Language*, 67:101151, 2021.